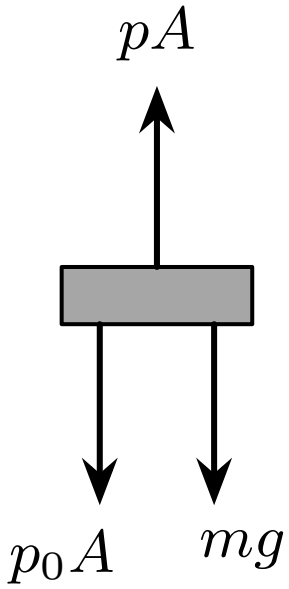
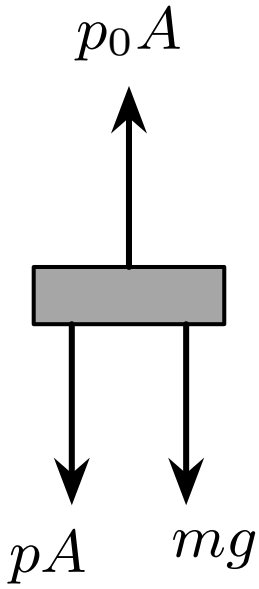


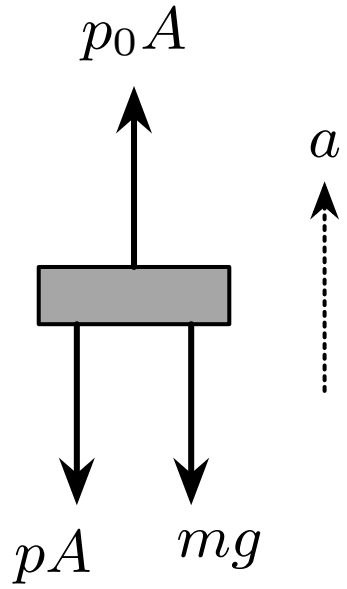
open end up, at rest

$pA = p_0A + mg$



open end down, at rest

$p_0A = pA + mg$



open end down, lift accelerating up at $\frac{1}{2}g$

$p_0A - pA - mg = \frac{1}{2}mg$